

End of Semester Presentation: Self-Adjusting Photolithography Lens

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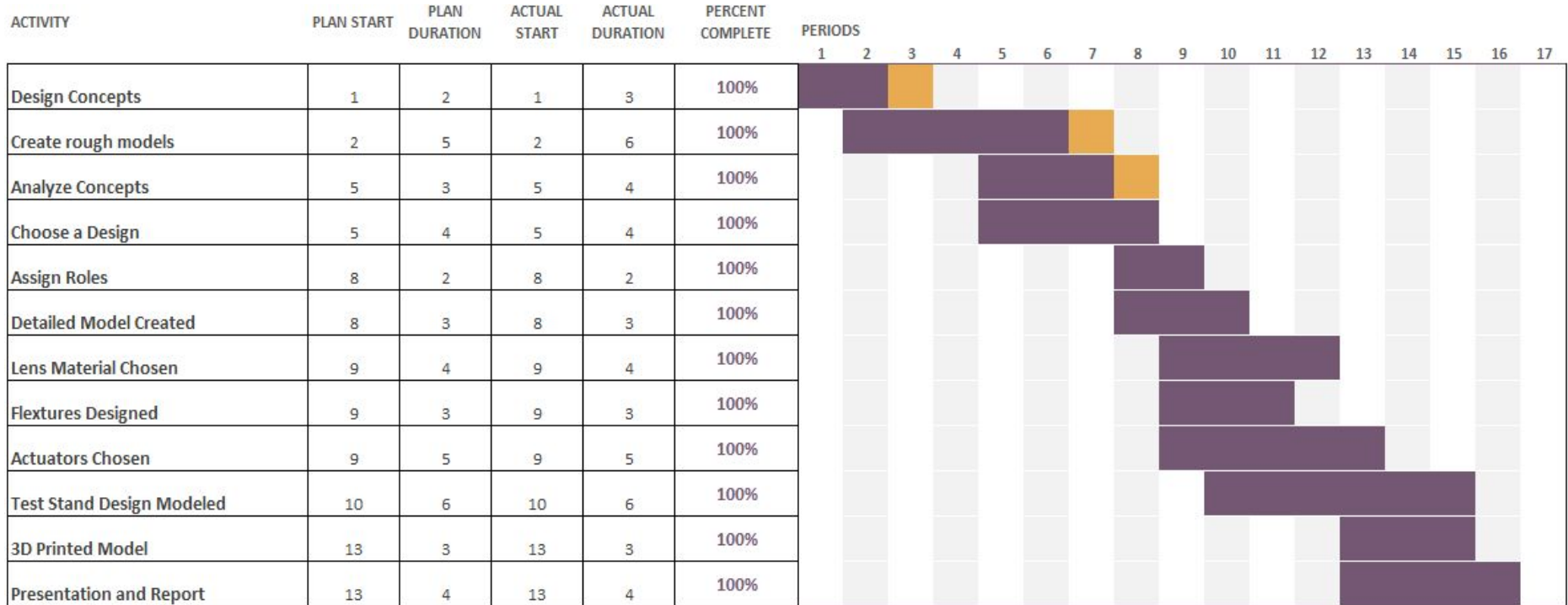
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Contents

- **Project Progress**
- **Project Description**
- **Initial and Final Actuation Designs**
- **Radial Compliant Solution**
- **Analysis of Design**
- **Future Steps**

Gantt Chart



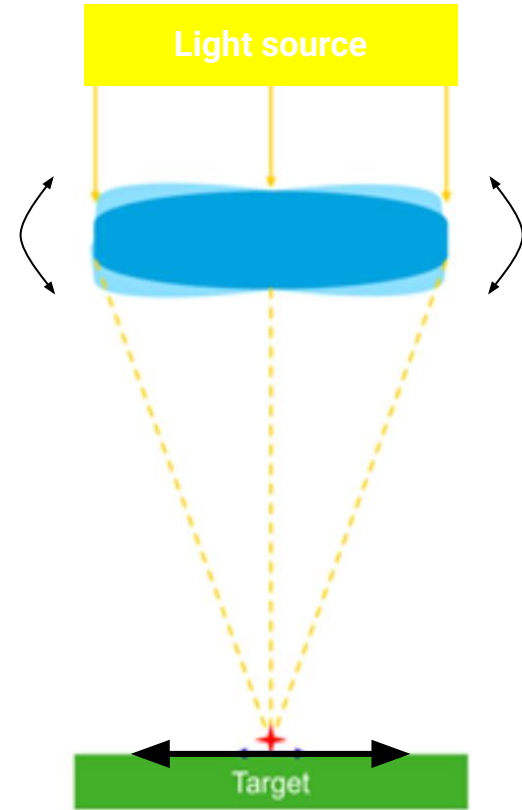
The precise control of light

(Background and motivation)

- 
- A diagram illustrating the precise control of light. It features a dark blue, upward-sloping surface on the right side. Three yellow lines with arrowheads at their ends originate from a point on this surface and extend upwards and to the left, representing reflected light rays. Additionally, three yellow lines with arrowheads at their ends originate from the same point on the surface and extend upwards and to the right, representing incident light rays. The background is a light blue gradient.
- In ASML's EUV Photolithography machine, an abundance of mirrors and lenses are implemented to precisely control the direction and use of light, including high energy ultraviolet light
 - Light is a tool used in many industries, from medical imaging and radiation treatments to consumer cameras
 - It is therefore important to find methods to control and direct light rays

Project Description/ Problem statement

- We must design a mechanical system to tilt an optic lens along two axes, to shift a focal point.
- The system should be self-adjusting to account for vibration and thermal disturbances.
- Have a radially compliant mounting-mechanism to reduce any stresses in the lens induced from thermal expansion.



Requirements and Specifications

Requirements

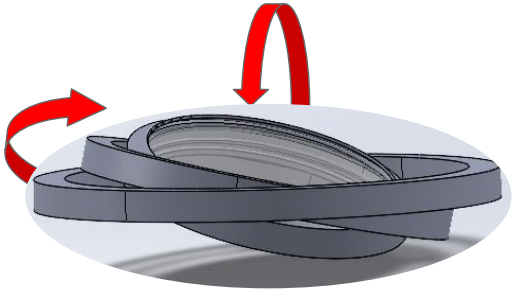
- Lens-tilting capacity: in the range of +/- 10 degrees in the Rx and Ry directions
- Movements in-plane: should have a 1000:1 ratio to lateral movements between planes

Design Parameters

- Lens optic: is specified between 4-8" Outer Diameter
- Mounting ring: between a 6" Inner diameter and a 10" OD
- Lens positioning: ideally occur in under 3 seconds

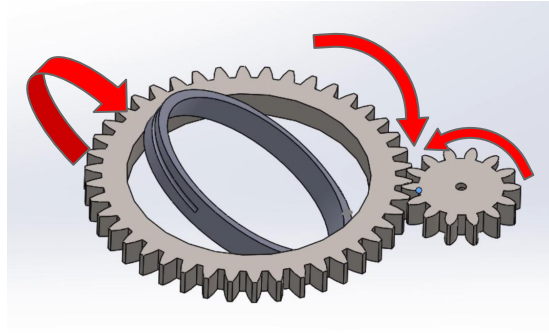
Actuation Design Concepts

1) Pitch and Roll



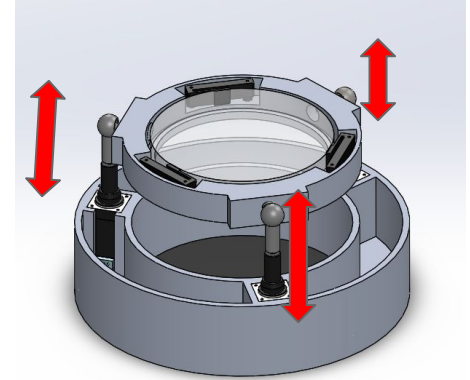
Tilt about X and Y axis.

2) Pitch and Yaw



Single tilt and rotator track

3) Linear actuation

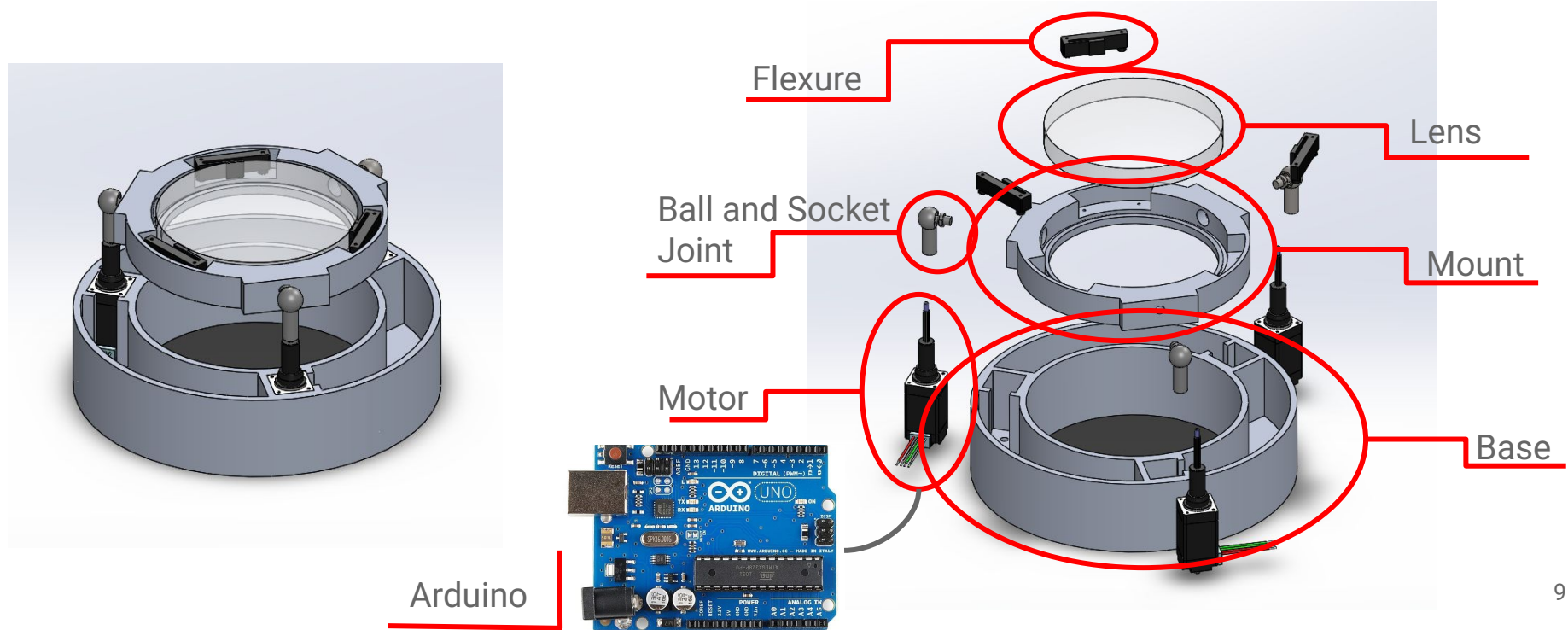


Translation on three joints

Final Decision and Reasoning

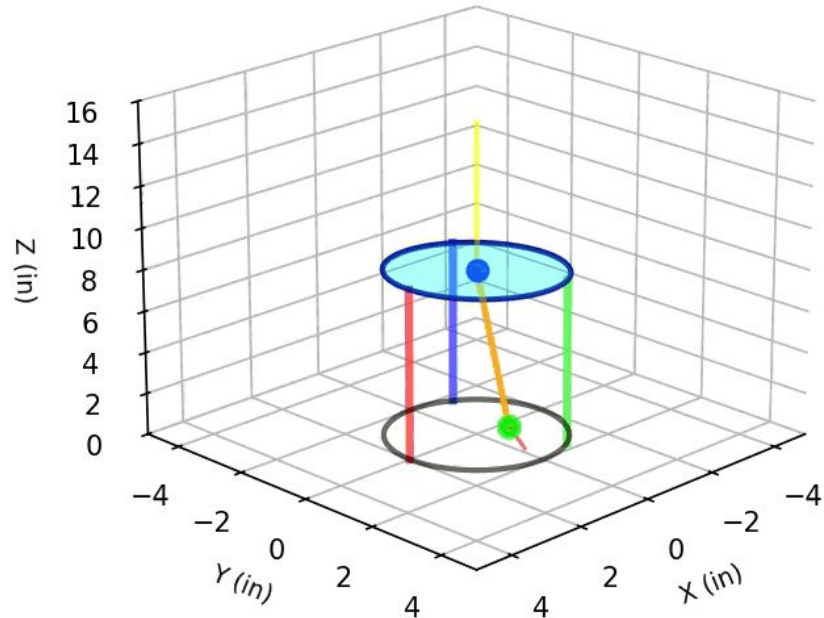
Design Attribute		Design 1		Design 2		Design 3	
	Weight Factor	Rating	Score	Rating	Score	Rating	Score
Speed of Operation	2	4	8	1	2	4	8
Accuracy	5	4	20	4	20	4	20
Size and and Weight	4	4	16	3	12	3.5	14
Environmental Robustness	1.5	3	4.5	2	3	4	6
Easy Assembly	5	3	15	3	15	4	20
Component Complexity	4	3	12	1	4	4	16
Environmental Impact	1	5	5	5	5	5	5
Life Cycle costs	3.5	3	10.5	2	7	4	14
Human Factors and Appearance	5	5	25	5	25	5	25
Manufactorablilty	5	3	15	3	15	4	20
Complexity of Calculation	1	3	3	3	3	1	1
Wiring	4	2	8	3	12	5	20
	Total Score		134		111		149

Exploded View of Linear Actuator Model (current design)

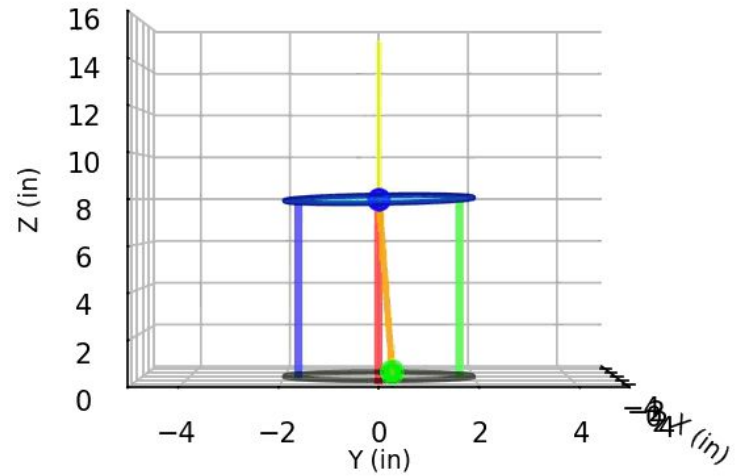


Simulation of Model in Motion

Isometric View



Side View (Y-axis)



Lens: Final Selection



Lens Specifications

- Double Convex Lens
- Diameter: 100 mm
- Center Thickness: 26.6 mm
- Edge Thickness: 13.8 mm
- Focal Length: 202 mm

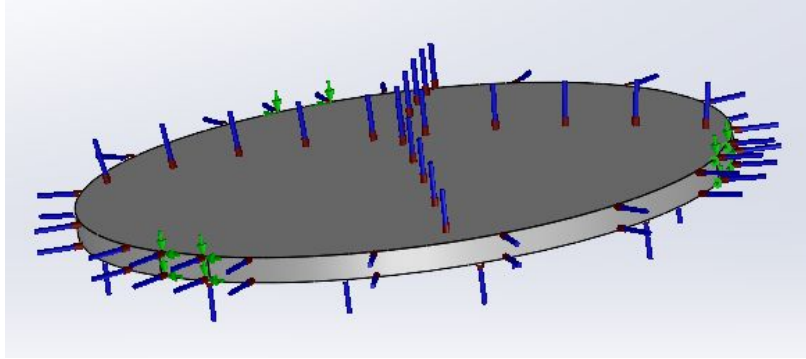
Design Considerations

- Lens Material: Glass
 - Refractive Index = $n = 1.5$
- Radius of Curvature: 198 mm
 - Used Thick Lens Equation

$$\frac{1}{f} = (n - 1) \left[\frac{1}{R} - \frac{1}{R} + \frac{(n - 1)d}{nR^2} \right]$$

$$R = \sqrt{\frac{(n - 1)^2 d \cdot f}{n}}$$

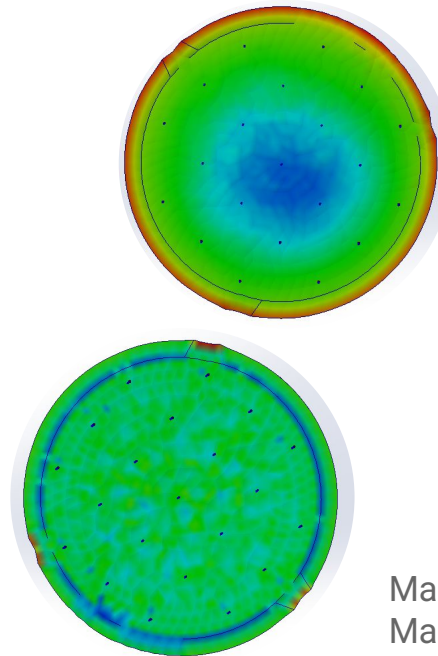
FEA Analysis 1: Thermal Expansion



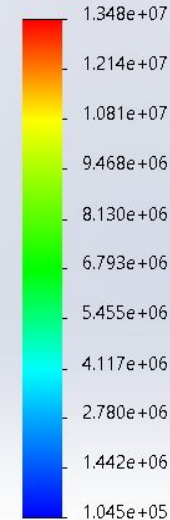
A 40C boundary condition is applied on all faces and A total normal elastic support of 710,061N/m is applied across all the flexures contact faces.

The lens must be fixed in shear on those 3 faces to be solvable, so stress at the contact points will be ignored

Results:

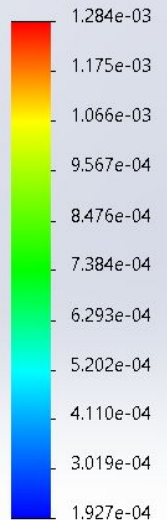


von Mises (N/m²)



→ Yield strength: 3.500e+07

URES (mm)

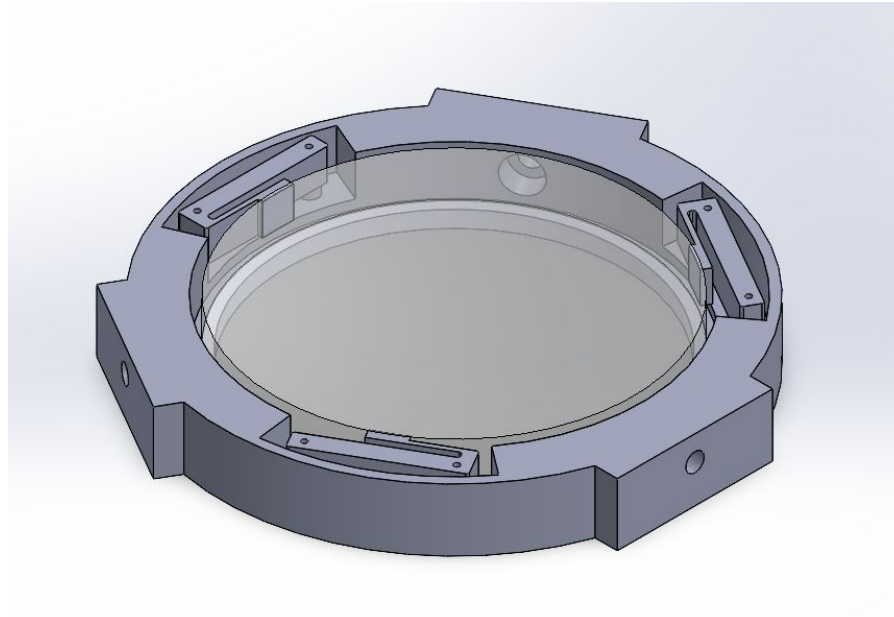


Max Displacement: 0.00128 mm

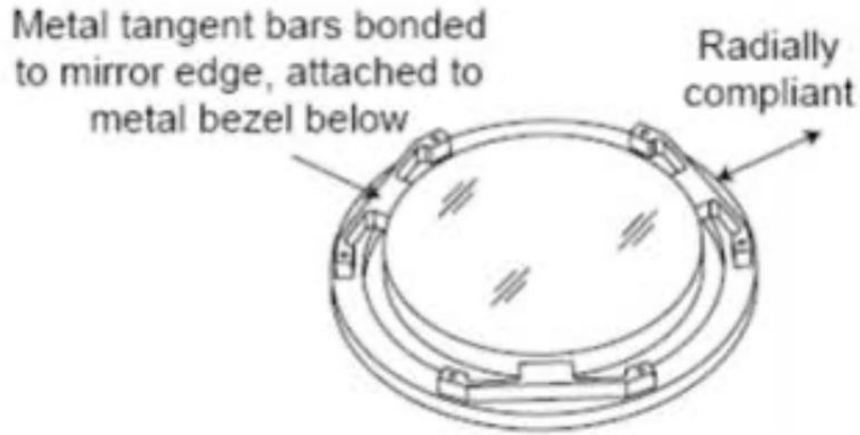
Max stress: ~8 MPa

Radial Compliant Mechanism: The Edge Bonded Flexure

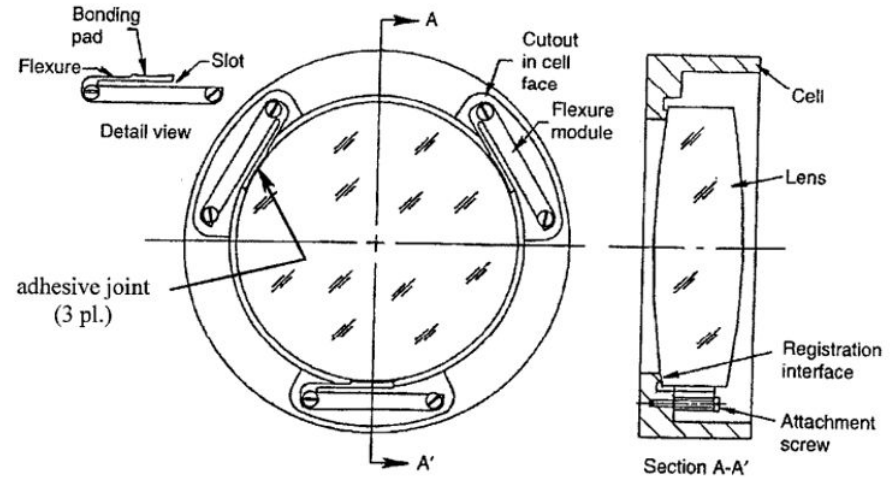
- Allows for the the thermal expansion of the lens while keeping the lens stiff in the Z-direction



Literature Search for Radially Compliant Mechanism



1) Yoder, Mounting Optics In Optical Instruments 2nd Ed.

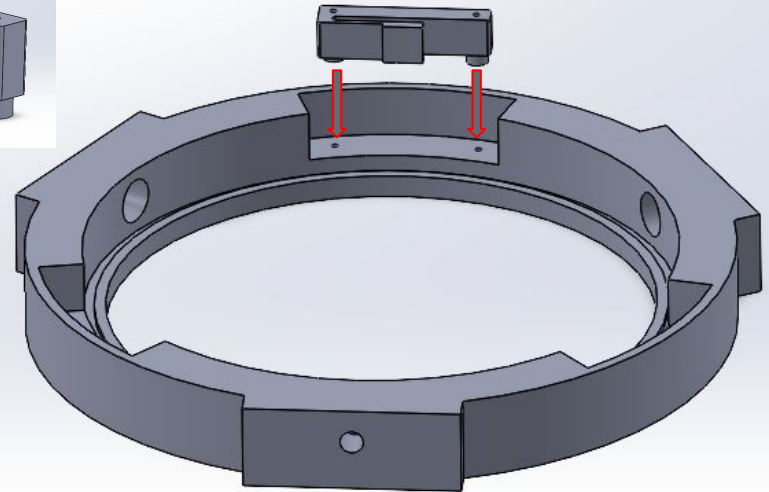
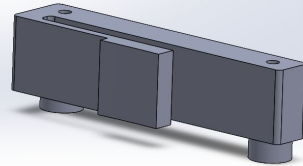


2) Ahmad and Huse (1990):
Patent Number: US-4929054-A

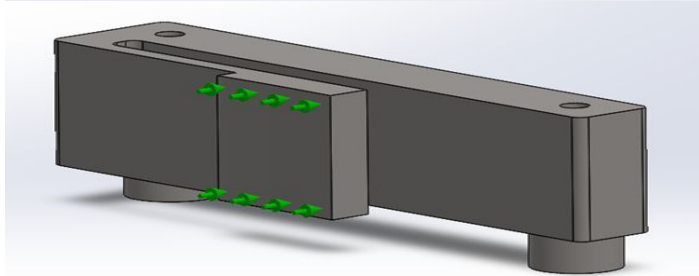
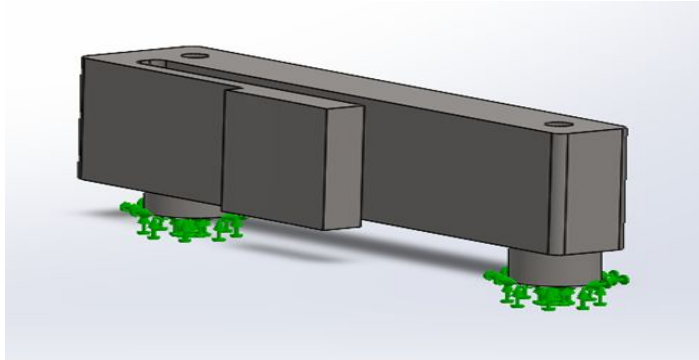
Radial Compliant Mechanism: Final Decision

- Flexures are connected on one side, allowing greater flexibility
- Flexures are removable, so the mount ring can be a lighter material
- The lens assembly process is made easier

	Design 1	Design 2
Design Attribute		
Flexibility	3	5
Mount Weight	2	4
Ease of Assembly	2	4
Total Score:	7	13

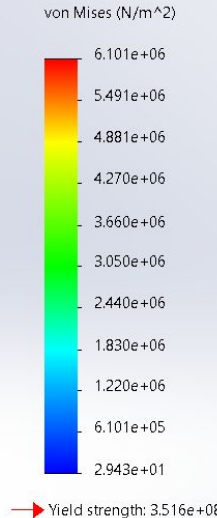
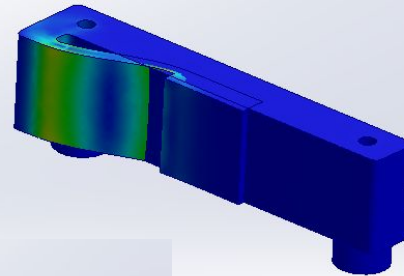
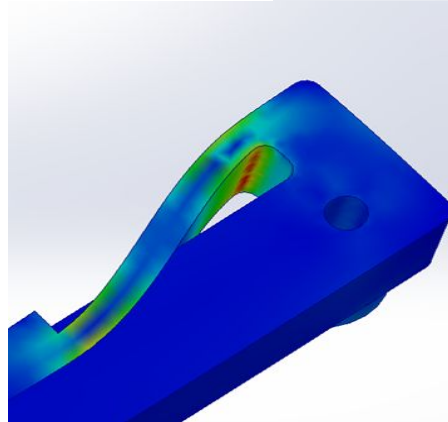


FEA Analysis 2: Edge Flexures



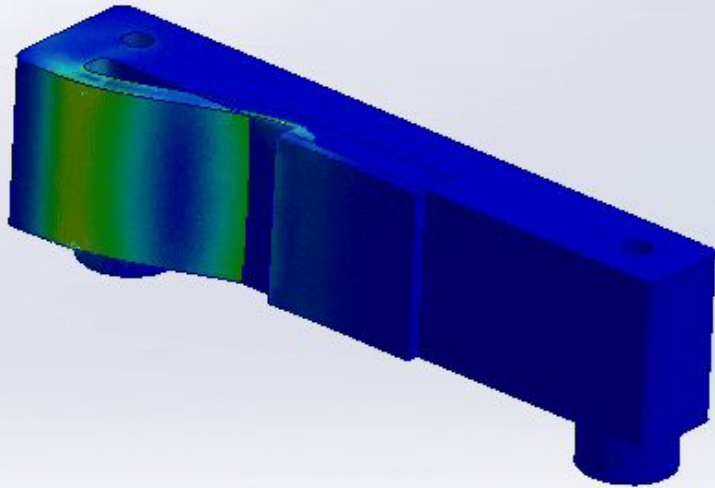
The flexure is fixed by the bottom legs and a prescribed displacement of 0.001284mm is applied to the flexure face.

Results:



Flexures have a static FOS of above 50

Fatigue Analysis for Flexure



$$\sigma_{max} = 6.101 \text{ MPa}; \quad \sigma_{min} = 0 \text{ MPa}$$

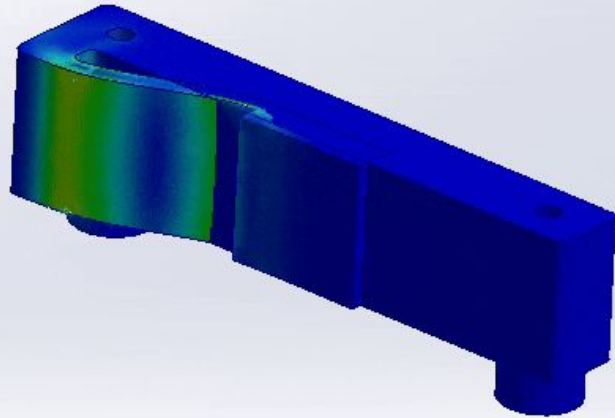
$$\sigma_m = \frac{6.101 \text{ MPa} + 0 \text{ MPa}}{2} = 3.0505 \text{ MPa}$$

$$\sigma_a = \sigma_m = 3.0505 \text{ MPa} \quad (\text{When } \sigma_{min} = 0)$$

From **Shigley's Mechanical Engineering Design 11th Edition**:

- **Stress Concentration Factor** (Figure A-15-6): $K_t = 1.4$
- **Notch Sensitivity Factor** (Figure 6-20): $q = 0.9$

Fatigue Analysis (Continued)



Notch Stress concentration:

$$K_f = 1 + q (K_t - 1)$$

$$K_f = 1 + 0.9(1.4 - 1) = 1.24$$

$$\sigma'_a = \sigma'_m = K_f * \sigma_{a \text{ or } m} = 1.24 * 3.0505 = 3.783 \text{ MPa}$$

Cycles: ($S_{ut} = 420 \text{ MPa}$)

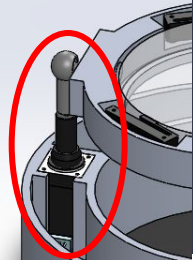
$$\sigma_{rev} = \frac{\sigma'_a}{1 - \left(\frac{\sigma'_m}{S_{ut}}\right)} = 3.817 \text{ MPa}$$

$$a = \frac{(f * S_{ut})^2}{S_e} = 680.4 \text{ MPa}; \quad b = -\frac{1}{3} \log \frac{f * S_{ut}}{S_e} = -.08509$$

$$N = \left(\frac{\sigma_{rev}}{a}\right)^{\frac{1}{b}} = 2.83 * 10^{26} \text{ Cycles}$$

Actuator for Lens Fixture

Goal: Identify a compact, precise, and stable actuator with enough stroke for optimal tilt.



Rack & Pinion

Converts rotational
→ linear motion

Issues:

- Backlash
- Tooth wear



Piezoelectric Motor

Uses ultrasonic vibrations
for extremely fine motion.

Issues:

- Out of Budget
- Does not hold position



Linear Stepper Motor

Lead screw converts motor
steps → linear motion.

Advantages:

- High precision & repeatability
- Holds position
- Compact form



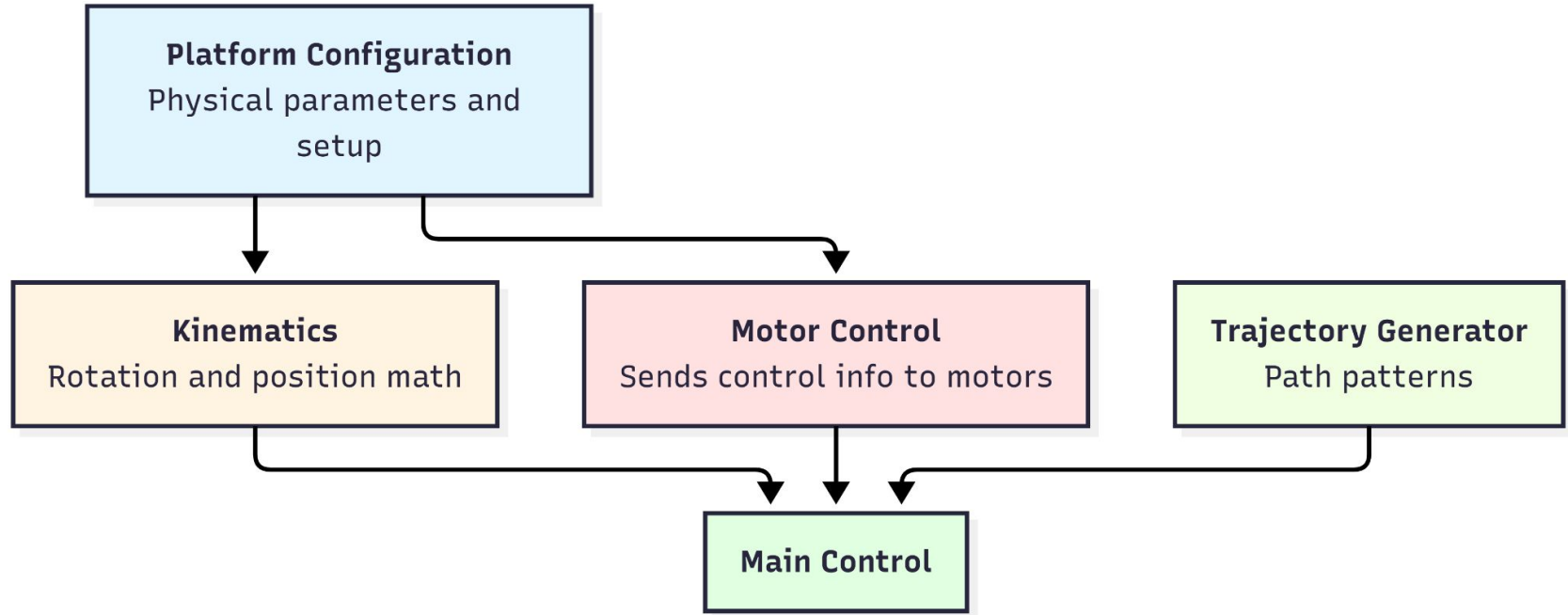
Linear actuator: Final selection

NEMA 8 Captive Linear Stepper Motor

- Compact 38.2mm length (20x20mm frame)
- 1.8 degree step angle
- 2 mm travel per revolution
- 0.01 mm moved per step
- Stroke of 19.1mm



Coding Outline



Model Testing

Tripod-Based Testing Mechanism

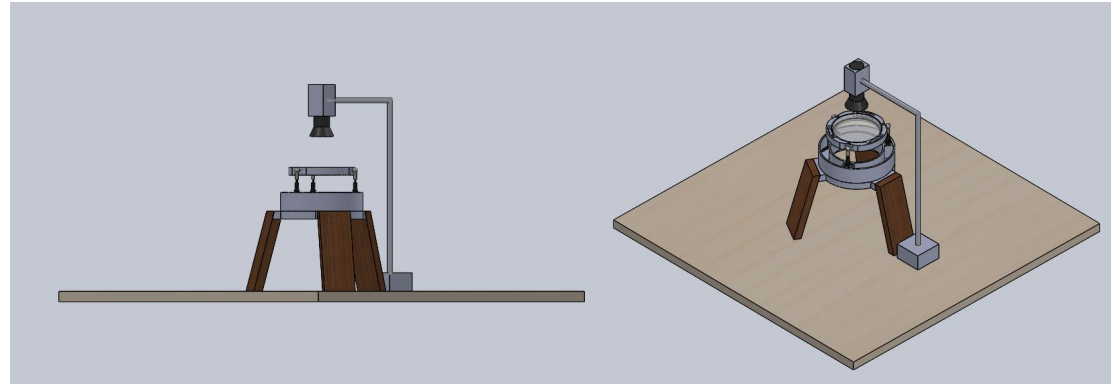
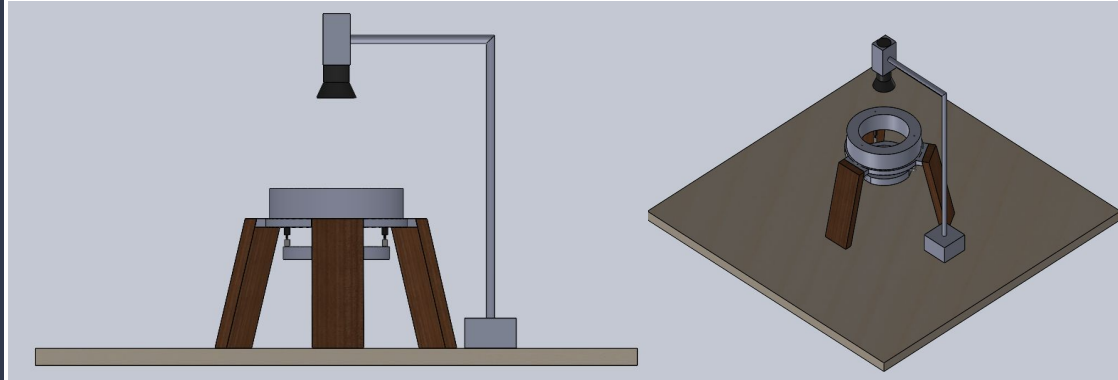
- Rigid tripod which will hold the assembly in stable testing position.

Two Mounting Configurations

- Pass Through Configuration
 - Lens Passes through the ring attached to tripod base.
- Resting Configuration
 - Plastic ring rests on the tripod ring, suspending the lens above the tripod.

Illumination Method

- Flashlight is mounted on the supporting structure, shining light directly into the lens.



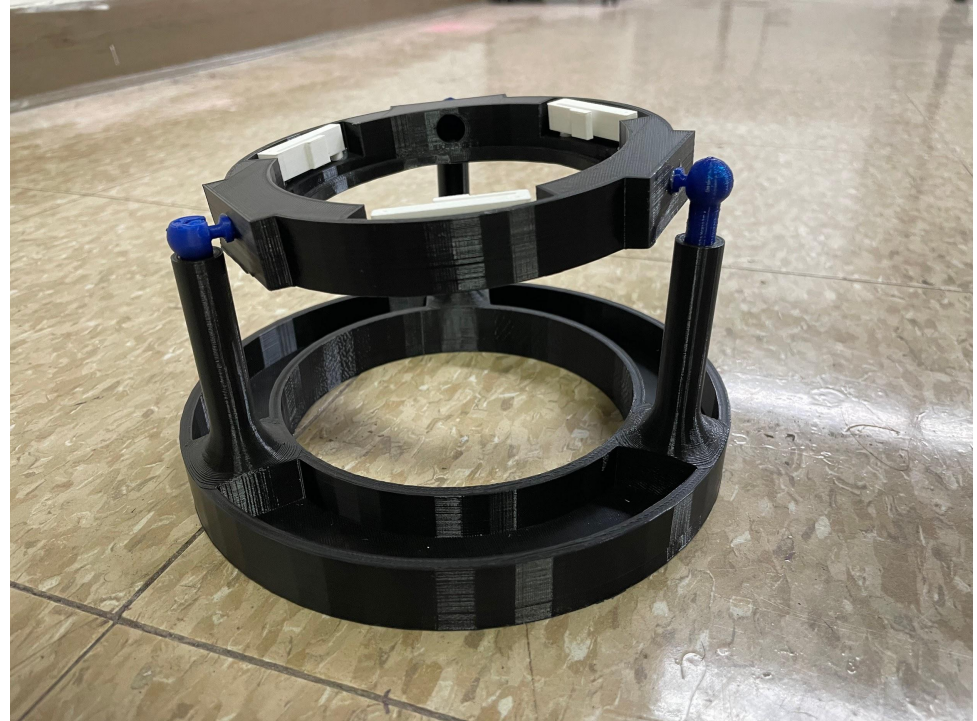
Summary & Conclusion

In this presentation:

- Project motivation and description:
 - Mechanical system to direct concentrated light
- Walked through our design process and decision to use a linear actuation model
- Walked through the design, selection and analysis of subsystems
 - Lens
 - Radially compliant end-flexure mechanism
 - Linear actuators
 - Testing apparatus for the final design.

Future work

- CNC real steel flexures
- Purchase selected components, the actuators and lens
- Produce remaining components and complete physical prototype
- Complete the design code
- Create a subsystem for the systems self adjusting requirement



End of Presentation!

Thank you,

The floor is open to questions

References

- 1) Yoder, Mounting Optics In Optical Instruments 2nd Ed.
- 2) Ahmad and Huse (1990): **Patent Number:**
US-4929054-A.
- 3) Richard G. Budynas, J. Keith Nisbett, Shigley's
Mechanical Engineering Design 11th Edition.